

MSCF Probability: Problem Session #2

The following exercises are taken from past exams in this course.

It is worth noting the grading guidelines that are utilized on the exam:

Grading Guidelines:

The following are reasons why you may lose points:

1. Failure to mention when an independence assumption is utilized.
2. Failure to mention when the assumption that random variables are uncorrelated is utilized.
3. Failure to specify clearly the support of a distribution.

You do **not** need to cite the names of rules and definitions when you use them, such as

1. The Rule for the Lazy Statistician
2. The Iterated Expectation Rule
3. The Rule for Conditional Probability
4. The Addition Rule
5. The Chain Rule
6. ...

But, in general, any work that does not make it clear how the answer was found is subject to losing points. In other words, even if you do not justify every step, there should be sufficient number of steps to make it clear you know how to do the derivation. Lack of clarity can also result from messy writing: write neatly.

1. X has the Exponential(λ) distribution, i.e. $E(X) = 1/\lambda$. What is $E(X^2)$? **(2)**
 $2/\lambda^2$

2. Which of the following are always true regarding $P(A|B)$? **(2)**
(CHOOSE AS MANY AS APPROPRIATE)

- (a) $P(A|B)$ is only defined if $P(B) > 0$.
- (b) $P(A|B) = P(A \cap B)/P(B)$ if $P(B) > 0$. **YES**
- (c) $P(A|B) \geq P(A)$ for any events A, B .
- (d) $P(A|B) = P(A)$ when A, B are independent events. **YES**

3. A fair coin is tossed twice. Define

$$\begin{aligned}A_1 &= \{\text{First toss is "heads"}\} \\A_2 &= \{\text{Second toss is "heads"}\} \\B &= \{\text{the two tosses have different outcomes}\}\end{aligned}$$

In other words, B is the event that exactly one of the two tosses is “heads.”

What can be said about these events? **(2)**
(CHOOSE AS MANY AS APPROPRIATE)

- (a) Events A_1 and A_2 are independent. **YES**
- (b) Events A_1 and A_2 are independent conditional on B .
- (c) Events A_1 and B are independent. **YES**

4. Suppose that “events” arrive as a Poisson process with rate $\lambda = 24$ per hour. What is the probability of zero events during a five minute period of time? **(2)**
 $\exp(-2)$

5. Suppose that X is a random variable and $Y = X + b$, where b is a constant. What is the relationship between the moment generating function (MGF) for X and the MGF for Y ? **(CHOOSE ONE) (2)**

- (a) $M_Y(t) = e^{tb}M_X(t)$ **CORRECT**
- (b) $M_Y(t) = M_X(tb)$
- (c) $M_Y(t) = M_X(t)$
- (d) $M_Y(t) = bM_X(t)$

6. A factory produces X_n gadgets on day n . It can be reasonably assumed that $X_1, X_2, \dots$ are independent and identically distributed random variables, each with mean 5 and variance 9. Approximate the probability that the number of gadgets produced in 100 days is fewer than 440. **(3)**

Use CLT. The desired probability is approximately $\Phi(-2) = 0.022$.

7. Suppose a fair, six-sided die is rolled repeatedly. Let X equal the number of rolls required in order to see each side (“1” thru “6”) at least once. What is the expected value of X ? **(3)**

Hint: Let X_1 be the number of rolls required to see the first “new” side, let X_2 equal the number of rolls required to see the second “new” side, and so forth.

Then, $X = X_1 + X_2 + X_3 + X_4 + X_5 + X_6$.

Using the hint, note that $X_i - 1$ is Geometric($(7 - i)/6$) and hence $E(X_i) = 6/(7 - i)$. Find that $E(X) = 14.7$.

8. Suppose a stick of length one is broken by choosing a point uniformly on 0 to 1. This creates two pieces, call them the “left” and “right” pieces. The “right” piece is discarded, and the “left” piece is then broken again at a position that is uniformly distributed along its length. Again, the “right” piece is discarded. Let X denote the length of the remaining piece.

- (a) What is the expected value of X ? **(3)**

Hint: Use iterated expectation.

Let Y denote the position of the first break. Then $E(X) = E(E(X|Y)) = E(0.5Y) = 0.5E(Y) = 0.25$.

- (b) Derive the CDF of X . **(2)**

For $0 < x \leq 1$, $F_X(x) = x(1 - \log(x))$.

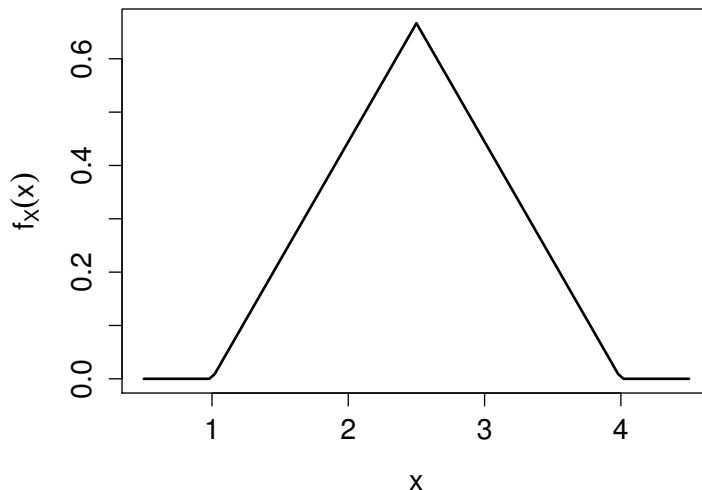
9. Suppose $a < b$. A random variable X is said to have the *triangular(a, b) distribution* if it has probability density function (pdf) of

$$f_X(x) = \begin{cases} \frac{2}{b-a} \left(1 - \frac{2}{b-a} \left| x - \frac{a+b}{2} \right| \right), & a < x < b \\ 0, & \text{otherwise} \end{cases}$$

The figure below shows this pdf for the case where $a = 1$ and $b = 4$.

Some other useful properties of this distribution: If X is triangular(a, b), then $E(X) = (a + b)/2$ and the moment generating function is, for $t \neq 0$,

$$M_X(t) = \left(\frac{e^{tb/2} - e^{ta/2}}{t(b-a)/2} \right)^2$$



- (a) Suppose that U_1 and U_2 are independent, and each has the $\text{Uniform}(a/2, b/2)$ distribution. Show that $U_1 + U_2$ has the $\text{triangular}(a, b)$ distribution. **(3)**

Follows directly from property of MGFs.

- (b) Suppose X has the $\text{triangular}(a, b)$ distribution. Does kX also have the triangular distribution for any choice of positive constant k ? Explain your answer clearly. **(3)**

By uniqueness of MGF, can see that kX would be $\text{triangular}(ka, kb)$.

10. A professor uses an automated cheating detection system. Of course, the system is not perfect. In fact, 10% of papers which are plagiarized are missed by the system, and 5% of papers which are not plagiarized are falsely reported as being plagiarized. Assume that 5% of all submitted papers will be plagiarized. If a randomly chosen paper is reported as being plagiarized, what is the probability that it actually was plagiarized? **(3)**

Note: A paper is plagiarized if it is copied from another source.

Use Bayes' Theorem. The desired probability is $18/37$.

11. Suppose that (X, Y) has joint density given by

$$f_{(X,Y)}(x, y) = \begin{cases} (9 - 6x - 6y + 4xy) / 4, & 0 \leq x < 1, 0 \leq y < 1 \\ 0, & \text{otherwise} \end{cases}$$

- (a) What is the marginal density for X ? **(3)**

$$f_X(x) = \begin{cases} 3/2 - x & 0 \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

(b) Are X and Y independent? Explain. **(2)**

Yes. By symmetry, we can see that the marginal for Y will be the same as that derived above. See that the product of the marginals returns the joint density.

12. Let $\mathbf{X}$ be multivariate normal: $\mathbf{X} = [X_1 \ X_2 \ X_3]^T \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with

$$\boldsymbol{\mu} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

and

$$\boldsymbol{\Sigma} = \begin{bmatrix} 1 & -0.3 & 0 \\ -0.3 & 1 & 0.2 \\ 0 & 0.2 & 2 \end{bmatrix}.$$

(a) What is the marginal distribution of X_1 ? **(3)**

Note: You not need show any derivations for this part, just state the answer.

X_1 is Normal with mean 1 and variance 1.

(b) What is the distribution of $2X_1 + X_2$? Fully specify the distribution, including the mean and variance. **(3)**

Normal with mean 0 and variance 3.8

13. Let X and Y be two random variables with joint probability density function (pdf) given by

$$f_{X,Y}(x, y) = \begin{cases} 2, & \text{if } x > 0, y > 0, \text{ and } x + y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Note that in this case, X and Y each have the Beta(1, 2) distribution.

What is the covariance between X and Y ? **(3)**

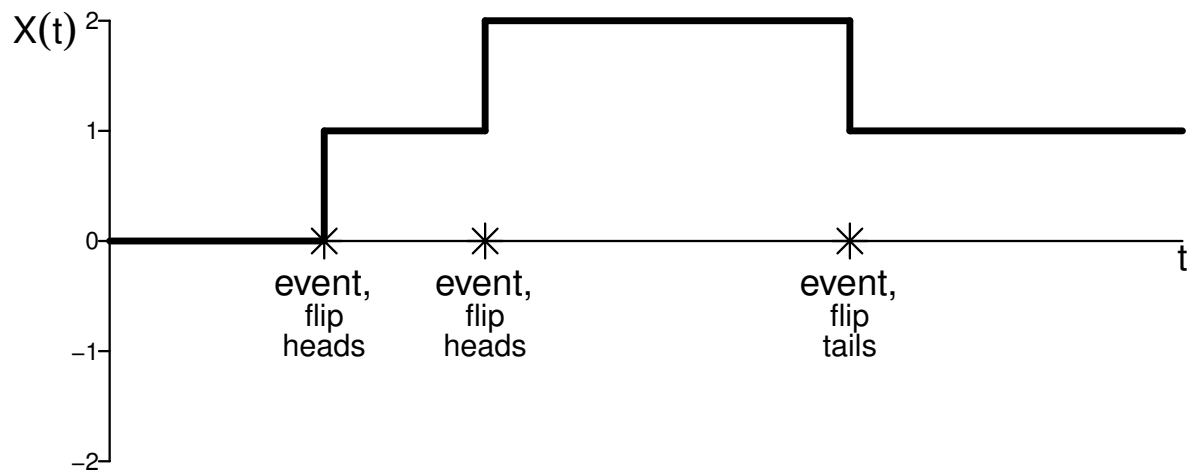
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14. Consider a Poisson process with rate λ . Each time an “event” occurs in this Poisson process, a fair coin is flipped to determine whether or not the stock price goes up (heads) or down (tails) by one unit. Let $X(t)$ denote the total price change up to time t .

The process starts at time $t = 0$, and $X(0) = 0$. The figure below illustrates an example of this process starting from time zero.

Define two additional random variables:

$$\begin{aligned} N(t) &= \text{the number of events that occurred during } (0, t] \\ U(t) &= \text{the number of “up steps” that occurred during } (0, t] \end{aligned}$$



Then, note that

$$\begin{aligned}
 X(t) &= \text{number of up steps during } (0, t] - \text{number of down steps during } (0, t] \\
 &= U(t) - (N(t) - U(t)) \\
 &= 2U(t) - N(t)
 \end{aligned}$$

Now, fix two times r, s such that $0 < r < s$.

It should be intuitive that $E(X(r)) = E(X(s)) = 0$.

Also, recall that $N(s)$ has the $\text{Poisson}(\lambda s)$ distribution.

What is the expected value of $U(r)$? **(3)**

$0.5\lambda r$

15. Suppose that $X_1, X_2, \dots, X_n$ are iid with the $\text{Exponential}(\lambda)$ distribution. Assume that $n > 100$. Then the Central Limit Theorem states that $\bar{X}$ will be approximately normal with mean $1/\lambda$ and variance $1/(n\lambda^2)$.

Suppose one wanted to estimate the tail probability $\tau = P(X_i > a)$ for some constant $a > 0$. Of course, since X_i is exponential,

$$\tau = \exp(-\lambda a).$$

Since λ is unknown we will replace λ with $1/\bar{X}$ and estimate this probability as

$$\hat{\tau} = \exp(-a/\bar{X})$$

Construct an appropriate approximation to the distribution of $\hat{\tau}$. **(3)**

Note: Your answer should only be in terms of λ , a , and n , but should otherwise completely specify the distribution.

Using the Delta method, we find that $\hat{\tau}$ is approximate normal with mean $\exp(-\lambda a)$ and variance $\exp(-2a\lambda)(\lambda^2 a^2/n)$.